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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 27th August Evening Shift

JEE Main Question Paper with Answer Key

27 Aug 2021	90	23	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	19 questions
Physics	26 questions
Out of syllabus	23 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

The addition of dilute NaOH to Cr^{3+} salt solution will give :

(A) a solution of $[Cr(OH)_4]^-$

(B) precipitate of $Cr_2O_3(H_2O)_n$

(C) precipitate of $[Cr(OH)_6]^{3-}$

(D) precipitate of $Cr(OH)_3$

Open solution

Q2. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : Ethyl pent-4-yn-oate on reaction with CH_3MgBr gives a 3°-alcohol.

Statement II : In this reaction one mole of ethyl pent-4-yn-oate utilizes two moles of CH_3MgBr.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false.

(B) Statement I is false but Statement II is true.

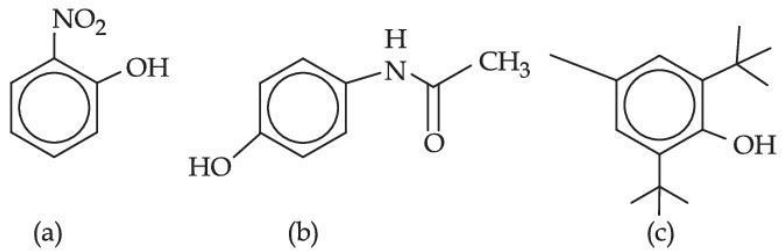
(C) Statement I is true but Statement II is false.

(D) Both Statement I and Statement II are true.

Open solution

Q3. MCQ (Single Correct Answer)

The compound/s which will show significant intermolecular H-bonding is/are :



(a) (b) (c)

(A) (b) only (B) (c) only (C) (a) and (b) only (D) (a), (b) and (c)

Open solution

Q4. MCQ (Single Correct Answer)

Which one of the following reactions will not yield propionic acid?

(A) $CH_3CH_2COCH_3 + OI^-/H_3O^+$

(B) $CH_3CH_2CH_3 + KMnO_4(Heat), OH^-/H_3O^+$

(C) $CH_3CH_2CCl_3 + OH^-/H_3O^+$

(D) $CH_3CH_2CH_2Br + Mg, CO_2$ dry ether/H_3O^+

Open solution

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Q5. MCQ (Single Correct Answer)

Open solution

The correct order of ionic radii for the ions, P^{3-}, S^{2-}, Ca^{2+}, K^{+}, Cl^{-} is :

(A) $P^{3-} > S^{2-} > Cl^{-} > K^{+} > Ca^{2+}$

(B) $Cl^{-} > S^{2-} > P^{3-} > Ca^{2+} > K^{+}$

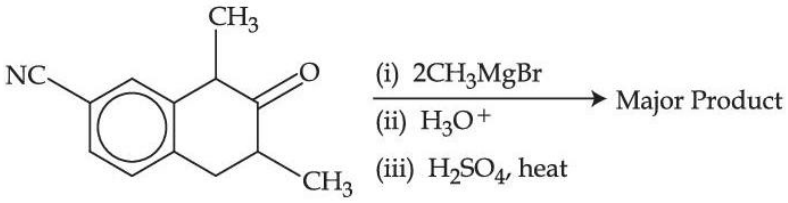
(C) $P^{3-} > S^{2-} > Cl^{-} > Ca^{2+} > K^{+}$

(D) $K^{+} > Ca^{2+} > P^{3-} > S^{2-} > Cl^{-}$

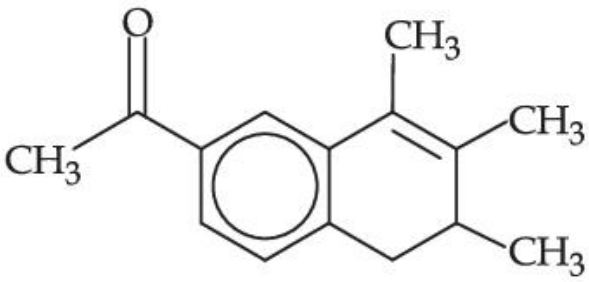
Q6. MCQ (Single Correct Answer)

Open solution

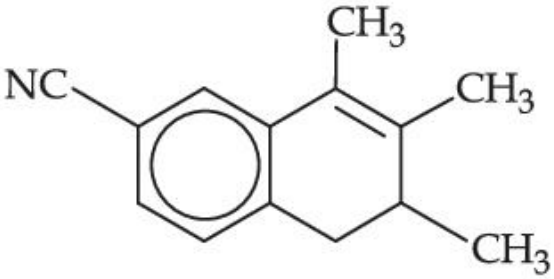
Which one of the following is the major product of the given reaction?



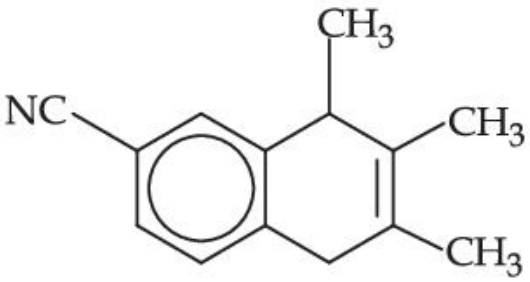
(A)



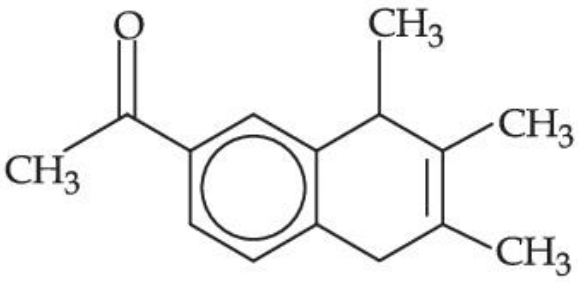
(B)



(C)



(D)

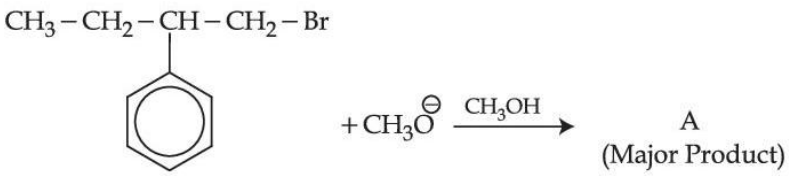


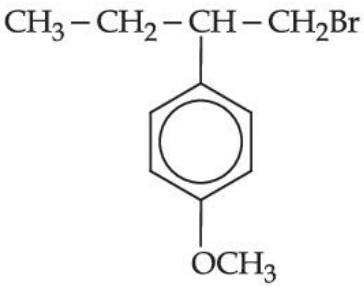
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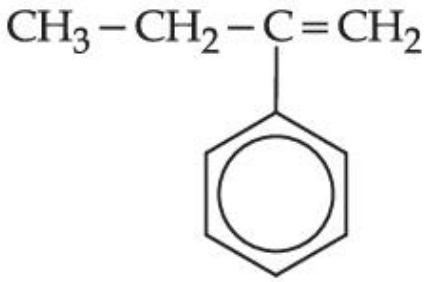
Q7. MCQ (Single Correct Answer)

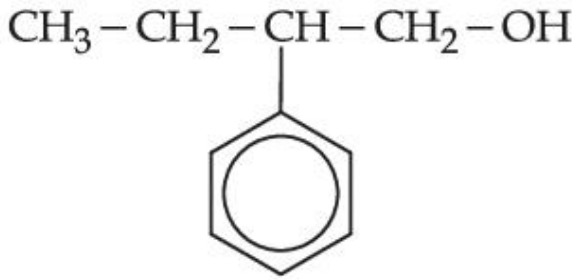
Open solution

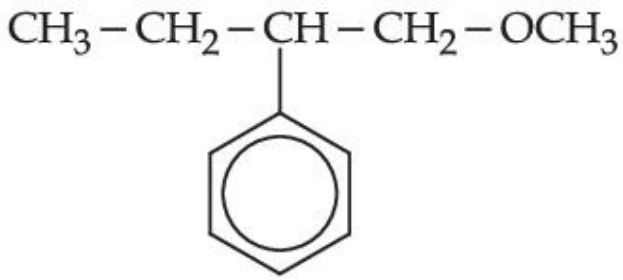
The major product (A) formed in the reaction given below is :



(A) 

(B) 

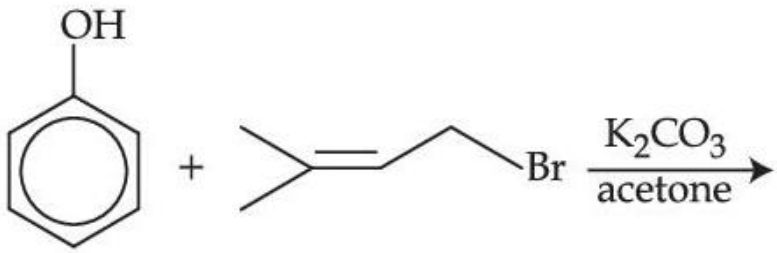
(C) 

(D) 

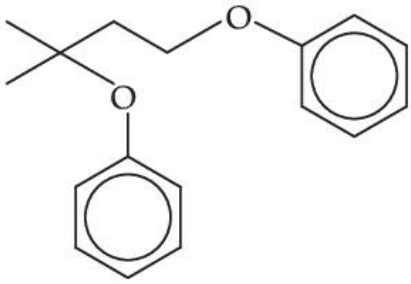
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Q8. MCQ (Single Correct Answer)

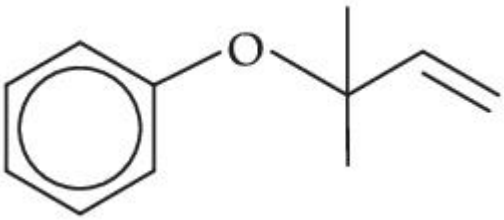
The major product of the following reaction, if it occurs by S_N2 mechanism is :



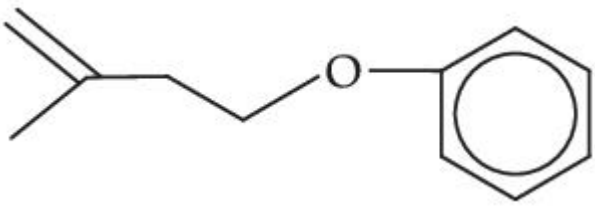
(A)



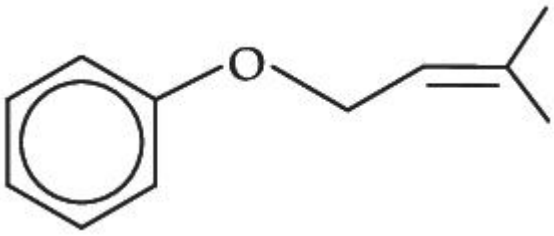
(B)



(C)



(D)



Open solution

Q9. MCQ (Single Correct Answer)

Potassium permanganate on heating at 513 K gives a product which is :

(A) paramagnetic and colourless

(B) diamagnetic and green

(C) diamagnetic and colourless

(D) paramagnetic and green

Open solution

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Q10. MCQ (Single Correct Answer)

Which one of the following tests used for the identification of functional groups in organic compounds does not use copper reagent?

Open solution

(A) Barfoed's test

(B) Seliwanoff's test

(C) Benedict's test

(D) Biuret test for peptide bond

Q11. MCQ (Single Correct Answer)

Hydrolysis of sucrose gives :

Open solution

(A) α-D-(−)-Glucose and β-D-(−)-Fructose

(B) α-D-(+)-Glucose and α-D-(−)-Fructose

(C) α-D-(−)-Glucose and α-D-(+)-Fructose

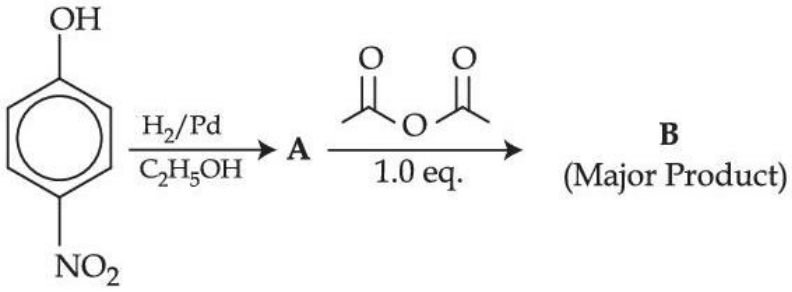
(D) α-D-(+)-Glucose and β-D-(−)-Fructose

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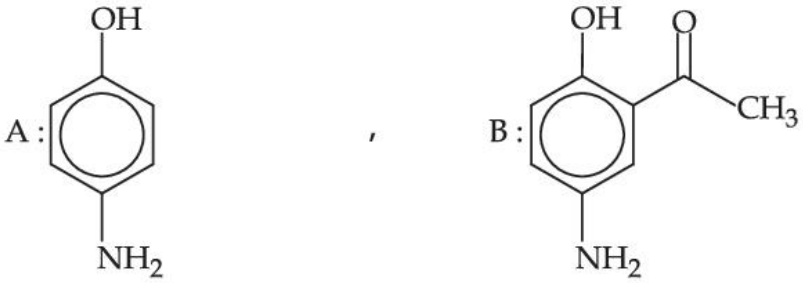
Q12. MCQ (Single Correct Answer)

Open solution

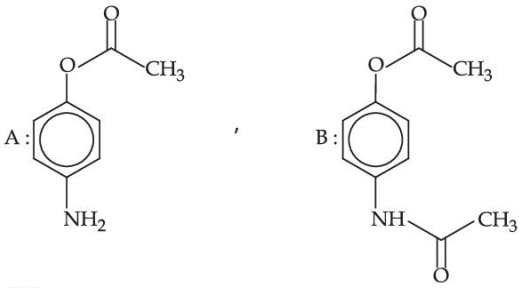
The correct structures of A and B formed in the following reactions are :



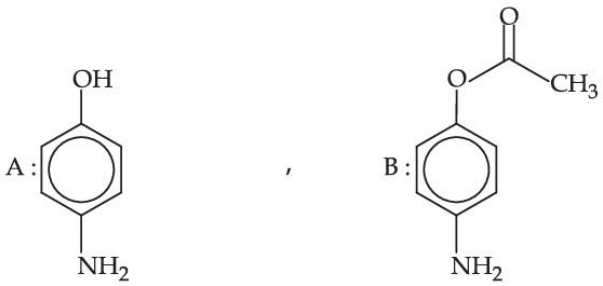
(A)



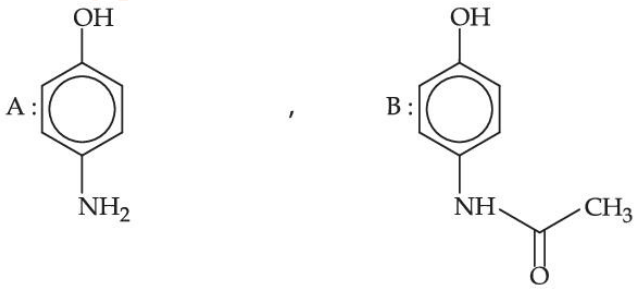
(B)



(C)



(D)



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Q13. Numerical

The first order rate constant for the decomposition of $CaCO_3$ at 700 K is $6.36 \times 10^{-3} s^{-1}$ and activation energy is 209 kJ mol^{-1}. Its rate constant (in s^{-1}) at 600 K is $x \times 10^{-6}$. The value of x is _____. (Nearest integer)

[Given R = 8.31 J K^{-1} mol^{-1}; log $6.36 \times 10^{-3} = -2.19$, $10^{-4.79} = 1.62 \times 10^{-5}$]

Open solution

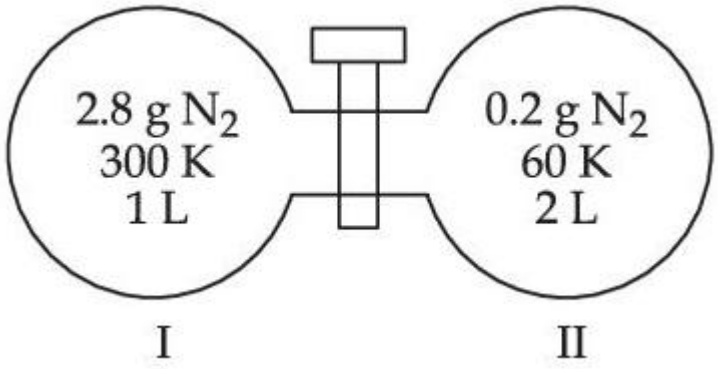
Q14. Numerical

The number of optical isomers possible for $[Cr(C_2O_4)_3]^{3-}$ is _____.

Open solution

Q15. Numerical

Two flasks I and II shown below are connected by a valve of negligible volume.



When the valve is opened, the final pressure of the system in bar is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

[Assume - Ideal gas; 1 bar = 10^5 Pa; Molar mass of N_2 = 28.0 g mol^{-1}; R = 8.31 J mol^{-1} K^{-1}]

Open solution

Q16. Numerical

100 g of propane is completely reacted with 1000 g of oxygen. The mole fraction of carbon dioxide in the resulting mixture is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)

[Atomic weight : H = 1.008; C = 12.00; O = 16.00]

Open solution

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Q17. Numerical

40 g of glucose (Molar mass = 180) is mixed with 200 mL of water. The freezing point of solution is _____ K. (Nearest integer) [Given : K_f = 1.86 K kg mol^{-1}; Density of water = 1.00 g cm^{-3}; Freezing point of water = 273.15 K]

Open solution

Q18. Numerical

The resistance of a conductivity cell with cell constant 1.14 cm^{-1}, containing 0.001 M KCl at 298 K is 1500 Ω. The molar conductivity of 0.001 M KCl solution at 298 K in S cm^2 mol^{-1} is _____. (Integer answer)

Open solution

Q19. Numerical

The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1 mW and wavelength of 1000 nm, in 0.1 second is $x \times 10^{13}$. The value of x is _____. (Nearest integer)

($h = 6.63 \times 10^{-34}$ Js, $c = 3.00 \times 10^8$ ms^{-1})

Open solution

Q20. Numerical

When 5.1 g of solid NH_4HS is introduced into a two litre evacuated flask at 27°C, 20% of the solid decomposes into gaseous ammonia and hydrogen sulphide. The K_p for the reaction at 27°C is $x \times 10^{-2}$. The value of x is _____. (Integer answer) [Given R = 0.082 L atm K^{-1} mol^{-}]

Open solution

Q21. Numerical

The number of species having non-pyramidal shape among the following is _____.

(A) SO_3

(B) NO_3^-

(C) PCl_3

(D) CO_3^{2-}

Open solution

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Q22. Numerical

Data given for the following reaction is as follows :

$FeO_{(s)} + C_{(graphite)} \rightarrow Fe_{(s)} + CO_{(g)}$

Open solution

Substance	ΔH° (kJ mol^{-1})	ΔS° (J mol^{-1} K^{-1})
$FeO_{(s)}$	-266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe_{(s)}$	0	27.28
$CO_{(g)}$	-110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is _____. (Integer answer)

Mathematics 19 questions

Q23. MCQ (Single Correct Answer)

The angle between the straight lines, whose direction cosines are given by the equations 2l + 2m − n = 0 and mn + nl + lm = 0, is :

Open solution

(A) $\frac{\pi}{2}$ (B) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$

(C) $\cos^{-1}\left(\frac{8}{9}\right)$ (D) $\frac{\pi}{3}$

Q24. MCQ (Single Correct Answer)

Let $A = \begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, where [t] denotes the greatest integer less than or equal to t. If det(A) = 192, then the set of values of x is the interval :

Open solution

(A) [68, 69) (B) [62, 63) (C) [65, 66) (D) [60, 61)

Q25. MCQ (Single Correct Answer)

Let M and m respectively be the maximum and minimum values of the function f(x) = $\tan^{-1}$ (sin x + cos x) in $\left[0, \frac{\pi}{2}\right]$, then the value of tan(M − m) is equal to :

Open solution

(A) $2 + \sqrt{3}$ (B) $2 - \sqrt{3}$ (C) $3 + 2\sqrt{2}$ (D) $3 - 2\sqrt{2}$

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Q26. MCQ (Single Correct Answer)

If two tangents drawn from a point P to the parabola $y^2 = 16(x - 3)$ are at right angles, then the locus of point P is :

(A) x + 3 = 0 (B) x + 1 = 0 (C) x + 2 = 0 (D) x + 4 = 0

Open solution

Q27. MCQ (Single Correct Answer)

If the solution curve of the differential equation $(2x - 10y^3)dy + ydx = 0$, passes through the points (0, 1) and (2, β), then β is a root of the equation :

(A) $y^5 - 2y - 2 = 0$ (B) $2y^5 - 2y - 1 = 0$ (C) $2y^5 - y^2 - 2 = 0$ (D) $y^5 - y^2 - 1 = 0$

Open solution

Q28. MCQ (Single Correct Answer)

Let $[\lambda]$ be the greatest integer less than or equal to λ. The set of all values of λ for which the system of linear equations

x + y + z = 4,

3x + 2y + 5z = 3,

$9x + 4y + (28 + [\lambda])z = [\lambda]$ has a solution is :

(A) R (B) $(-\infty, -9) \cup (-9, \infty)$

(C) $[-9, -8)$ (D) $(-\infty, -9) \cup [-8, \infty)$

Open solution

Q29. MCQ (Single Correct Answer)

The set of all values of k > −1, for which the equation $(3x^2 + 4x + 3)^2 - (k + 1)(3x^2 + 4x + 3)(3x^2 + 4x + 2) + k(3x^2 + 4x + 2)^2 = 0$ has real roots, is :

(A) $\left(1, \frac{5}{2}\right]$ (B) [2, 3)

(C) $\left[-\frac{1}{2}, 1\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right] - \{1\}$

Open solution

Q30. MCQ (Single Correct Answer)

A box open from top is made from a rectangular sheet of dimension a × b by cutting squares each of side x from each of the four corners and folding up the flaps. If the volume of the box is maximum, then x is equal to :

(A) $\frac{a+b-\sqrt{a^2+b^2-ab}}{12}$ (B) $\frac{a+b-\sqrt{a^2+b^2+ab}}{6}$

(C) $\frac{a+b-\sqrt{a^2+b^2-ab}}{6}$ (D) $\frac{a+b+\sqrt{a^2+b^2+ab}}{6}$

Open solution

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Q31. MCQ (Single Correct Answer)

Open solution

Let Z be the set of all integers,

$A = \{(x, y) \in Z \times Z : (x-2)^2 + y^2 \leq 4\}$

$B = \{(x, y) \in Z \times Z : x^2 + y^2 \leq 4\}$

$C = \{(x, y) \in Z \times Z : (x-2)^2 + (y-2)^2 \leq 4\}$

If the total number of relation from A ∩ B to A ∩ C is 2^p, then the value of p is :

(A) 16 (B) 25 (C) 49 (D) 9

Q32. MCQ (Single Correct Answer)

Open solution

The area of the region bounded by the parabola $(y-2)^2 = (x-1)$, the tangent to it at the point whose ordinate is 3 and the x-axis is :

(A) 9 (B) 10 (C) 4 (D) 6

Q33. MCQ (Single Correct Answer)

Open solution

If $y(x) = \cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x \in \left(\frac{\pi}{2}, \pi\right)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is :

(A) $-\frac{1}{2}$ (B) -1 (C) $\frac{1}{2}$ (D) 0

Q34. MCQ (Single Correct Answer)

Open solution

The value of the integral $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ is :

(A) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{6}\right)$

(C) $\frac{\pi}{8}\left(1-\frac{\sqrt{3}}{6}\right)$ (D) $\frac{\pi}{4}\left(1-\frac{\sqrt{3}}{2}\right)$

Q35. MCQ (Single Correct Answer)

Open solution

If $\lim_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right) = b$, then the ordered pair (a, b) is :

(A) $\left(1, \frac{1}{2}\right)$ (B) $\left(1, -\frac{1}{2}\right)$

(C) $\left(-1, \frac{1}{2}\right)$ (D) $\left(-1, -\frac{1}{2}\right)$

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Q36. Numerical

The probability distribution of random variable X is given by :

X	1	2	3	4	5
P(X)	K	2K	2K	3K	K

Let p = P(1 < X < 4 | X < 3). If 5p = λK, then λ equal to _____.

Open solution

Q37. Numerical

Let z_1 and z_2 be two complex numbers such that $\arg(z_1 - z_2) = \frac{\pi}{4}$ and z_1, z_2 satisfy the equation | z − 3 | = Re(z). Then the imaginary part of $z_1 + z_2$ is equal to _____.

Open solution

Q38. Numerical

Let S = {1, 2, 3, 4, 5, 6, 9}. Then the number of elements in the set T = {A $\subseteq$ S : A $\neq \phi$ and the sum of all the elements of A is not a multiple of 3} is _____.

Open solution

Q39. Numerical

$3 \times 7^{22} + 2 \times 10^{22} - 44$ when divided by 18 leaves the remainder _____.

Open solution

Q40. Numerical

An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to _____.

Open solution

Q41. Numerical

If $\int \frac{2e^x+3e^{-x}}{4e^x+7e^{-x}}dx = \frac{1}{14}(ux + v\log_e(4e^x + 7e^{-x})) + C$, where C is a constant of integration, then u + v is equal to _____.

Open solution

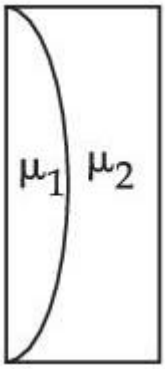
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Physics 26 questions

Q42. MCQ (Single Correct Answer)

Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.

Open solution

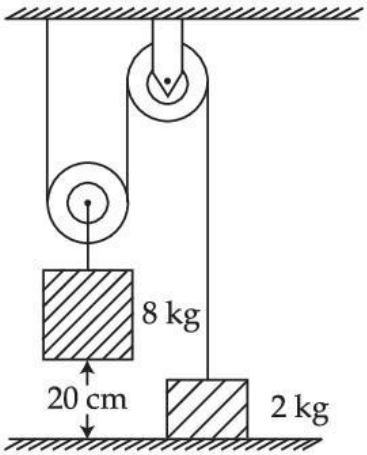


(A) $\frac{1}{\mu_2-\mu_1}$ (B) $\mu_1-\mu_2$ (C) $\frac{1}{\mu_1-\mu_2}$ (D) $\mu_2-\mu_1$

Q43. MCQ (Single Correct Answer)

The boxes of masses 2 kg and 8 kg are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 kg to strike the ground starting from rest. (use g = 10 m/s^2)

Open solution



(A) 0.34 s (B) 0.2 s (C) 0.25 s (D) 0.4 s

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Q44. MCQ (Single Correct Answer)

Water drops are falling from a nozzle of a shower onto the floor, from a height of 9.8 m. The drops fall at a regular interval of time. When the first drop strikes the floor, at that instant, the third drop begins to fall. Locate the position of second drop from the floor when the first drop strikes the floor.

Open solution

(A) 4.18 m (B) 2.94 m (C) 2.45 m (D) 7.35 m

Q45. MCQ (Single Correct Answer)

Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by :

Open solution

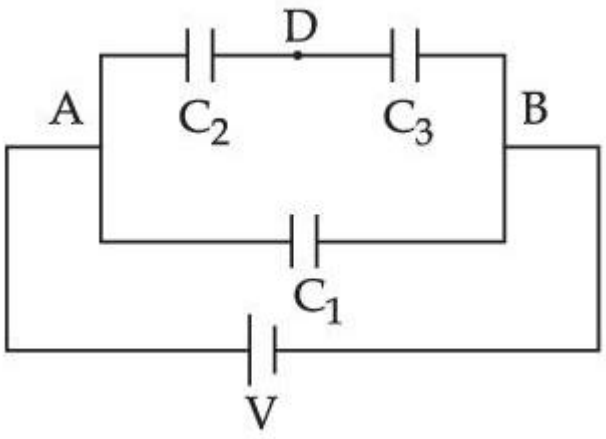
(A) $\frac{I_1 I_2}{(I_1+I_2)}(\omega_1-\omega_2)^2$ (B) $\frac{(I_1-I_2)^2\omega_1\omega_2}{2(I_1+I_2)}$

(C) $\frac{I_1 I_2}{2(I_1+I_2)}(\omega_1-\omega_2)^2$ (D) $\frac{(\omega_1-\omega_2)^2}{2(I_1+I_2)}$

Q46. MCQ (Single Correct Answer)

Three capacitors $C_1 = 2\mu F$, $C_2 = 6\mu F$ and $C_3 = 12\mu F$ are connected as shown in figure. Find the ratio of the charges on capacitors C_1, C_2 and C_3 respectively :

Open solution



(A) 2 : 1 : 1 (B) 2 : 3 : 3 (C) 1 : 2 : 2 (D) 3 : 4 : 4

Q47. MCQ (Single Correct Answer)

if the rms speed of oxygen molecules at 0°C is 160 m/s, find the rms speed of hydrogen molecules at 0°C.

Open solution

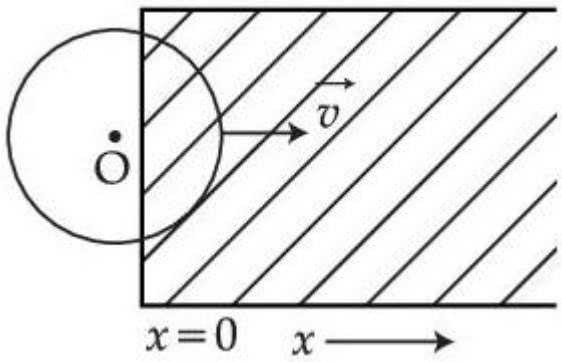
(A) 640 m/s (B) 40 m/s (C) 80 m/s (D) 332 m/s

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Q48. MCQ (Single Correct Answer)

A constant magnetic field of 1T is applied in the x > 0 region. A metallic circular ring of radius 1m is moving with a constant velocity of 1 m/s along the x-axis. At t = 0s, the centre of O of the ring is at x = −1m. What will be the value of the induced emf in the ring at t = 1s? (Assume the velocity of the ring does not change.)

Open solution



(A) 1V (B) 2πV (C) 2V (D) 0V

Q49. MCQ (Single Correct Answer)

A mass of 50 kg is placed at the centre of a uniform spherical shell of mass 100 kg and radius 50 m. If the gravitational potential at a point, 25 m from the centre is V kg/m. The value of V is :

(A) −60 G (B) +2 G (C) −20 G (D) −4 G

Open solution

Q50. MCQ (Single Correct Answer)

For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be :

(A) 1Ω (B) 5Ω (C) 4Ω (D) 2Ω

Open solution

Q51. MCQ (Single Correct Answer)

A monochromatic neon lamp with wavelength of 670.5 nm illuminates a photo-sensitive material which has a stopping voltage of 0.48 V. What will be the stopping voltage if the source light is changed with another source of wavelength of 474.6 nm?

(A) 0.96 V (B) 1.25 V (C) 0.24 V (D) 1.5 V

Open solution

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Q52. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

	List - I		List - II
(a)	R_H (Rydberg constant)	(i)	$kg\,m^{-1}s^{-1}$
(b)	h (Planck's constant)	(ii)	$kg\,m^2s^{-1}$
(c)	μ_B (Magnetic field energy density)	(iii)	m^{-1}
(d)	η (coefficient of viscocity)	(iv)	$kg\,m^{-1}s^{-2}$

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(B) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(C) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(D) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

Q53. MCQ (Single Correct Answer)

Open solution

If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density :

(A) $[FL^{-4}T^2]$ (B) $[FL^{-3}T^2]$ (C) $[FL^{-5}T^2]$ (D) $[FL^{-3}T^3]$

Q54. MCQ (Single Correct Answer)

Open solution

A coaxial cable consists of an inner wire of radius 'a' surrounded by an outer shell of inner and outer radii 'b' and 'c' respectively. The inner wire carries an electric current i_0, which is distributed uniformly across cross-sectional area. The outer shell carries an equal current in opposite direction and distributed uniformly. What will be the ratio of the magnetic field at a distance x from the axis when (i) x < a and (ii) a < x < b ?

(A) $\frac{x^2}{a^2}$ (B) $\frac{a^2}{x^2}$ (C) $\frac{x^2}{b^2-a^2}$ (D) $\frac{b^2-a^2}{x^2}$

Q55. MCQ (Single Correct Answer)

Open solution

The height of Victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?

[Given 1 cal = 4.2 J and specific heat of water = 1 cal $g^{-1}\,{}^\circ C^{-1}$]

(A) 0.147° C (B) 14.76° C (C) 1.476° C (D) 0.014° C

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Q56. MCQ (Single Correct Answer)

A player kicks a football with an initial speed of 25 ms^{-1} at an angle of 45° from the ground. What are the maximum height and the time taken by the football to reach at the highest point during motion ? (Take g = 10 ms^{-2})

Open solution

(A) h_{max} = 10 m
T = 2.5 s

(B) h_{max} = 15.625 m
T = 3.54 s

(C) h_{max} = 15.625 m
T = 1.77 s

(D) h_{max} = 3.54 m
T = 0.125 s

Q57. MCQ (Single Correct Answer)

The light waves from two coherent sources have same intensity $I_1 = I_2 = I_0$. In interference pattern the intensity of light at minima is zero. What will be the intensity of light at maxima?

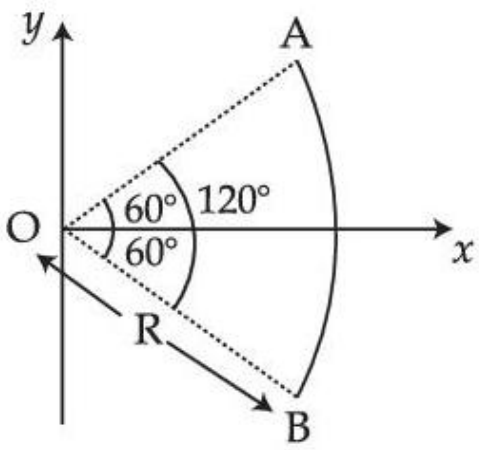
Open solution

(A) I_0 (B) 2 I_0 (C) 5 I_0 (D) 4 I_0

Q58. MCQ (Single Correct Answer)

Figure shows a rod AB, which is bent in a 120° circular arc of radius R. A charge (−Q) is uniformly distributed over rod AB. What is the electric field $\vec{E}$ at the centre of curvature O ?

Open solution



(A) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$
(B) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(\hat{i})$
(C) $\frac{3\sqrt{3}Q}{16\pi^2\varepsilon_0 R^2}(\hat{i})$
(D) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(-\hat{i})$

Q59. Numerical

Two simple harmonic motion, are represented by the equations $y_1 = 10\sin\left(3\pi t + \frac{\pi}{3}\right)$ $y_2 = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$ Ratio of amplitude of y_1 to y_2 = x : 1. The value of x is _____.

Open solution

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Q60. Numerical

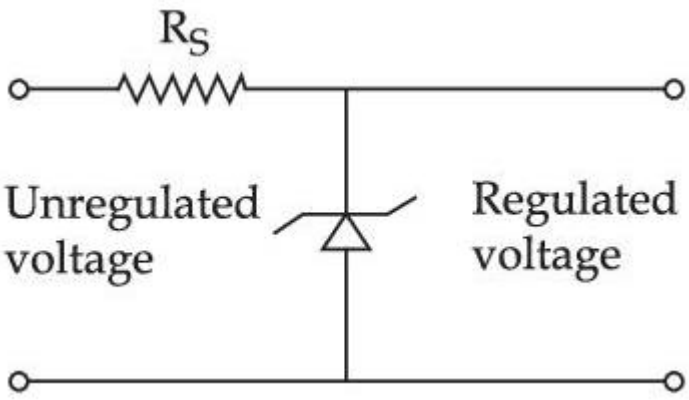
X different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to states with principal quantum number n = 6 ? The value of X is _____.

Open solution

Q61. Numerical

A zener diode of power rating 2W is to be used as a voltage regulator. If the zener diode has a breakdown of 10 V and it has to regulate voltage fluctuated between 6 V and 14 V, the value of R_s for safe operation should be _____ Ω.

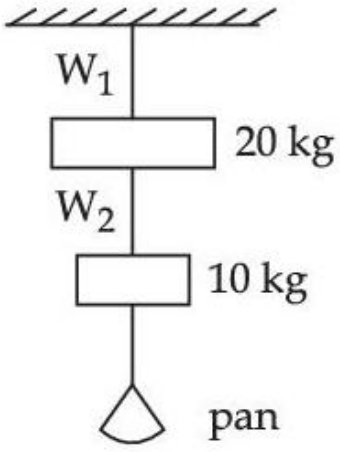
Open solution



Q62. Numerical

Wires W_1 and W_2 are made of same material having the breaking stress of 1.25×10^9 N/m^2. W_1 and W_2 have cross-sectional area of 8×10^{-7} m^2 and 4×10^{-7} m^2, respectively. Masses of 20 kg and 10 kg hang from them as shown in the figure. The maximum mass that can be placed in the pan without breaking the wires is _____ kg. (Use g = 10 m/s^2)

Open solution

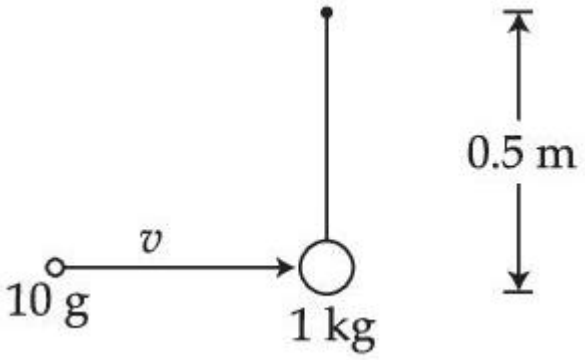


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Q63. Numerical

A bullet of 10 g, moving with velocity v, collides head-on with the stationary bob of a pendulum and recoils with velocity 100 m/s. The length of the pendulum is 0.5 m and mass of the bob is 1 kg. The minimum value of v = _____ m/s so that the pendulum describes a circle. (Assume the string to be inextensible and g = 10 m/s^2)

Open solution



Q64. Numerical

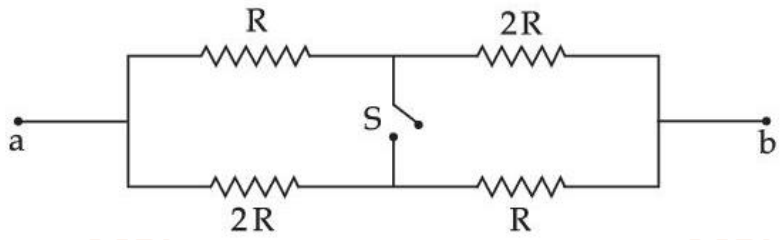
An ac circuit has an inductor and a resistor of resistance R in series, such that X_L = 3R. Now, a capacitor is added in series such that X_C = 2R. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5} : x$. The value of x is _____.

Open solution

Q65. Numerical

The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is x : 8. The value of x is _____.

Open solution



Q66. Numerical

A plane electromagnetic wave with frequency of 30 MHz travels in free space. At particular point in space and time, electric field is 6 V/m. The magnetic field at this point will be x $\times$ 10^{-8} T. The value of x is _____.

Open solution

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Q67. Numerical

A tuning fork is vibrating at 250 Hz. The length of the shortest closed organ pipe that will resonate with the tuning fork will be _____ cm. (Take speed of sound in air as 340 ms^{-1})

Open solution

Out of syllabus 23 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q68. MCQ (Single Correct Answer) · **Chemistry**

Choose the correct statement from the following :

(A) The standard enthalpy of formation for alkali metal bromides becomes less negative on descending the group.

(B) The low solubility of CsI in water is due to its high lattice enthalpy.

(C) Among the alkali metal halides, LiF is least soluble in water.

(D) LiF has least negative standard enthalpy of formation among alkali metal fluorides.

Open solution

Q69. MCQ (Single Correct Answer) · **Chemistry**

In stratosphere most of the ozone formation is assisted by :

(A) cosmic rays

(B) γ-rays

(C) ultraviolet radiation

(D) visible radiations

Open solution

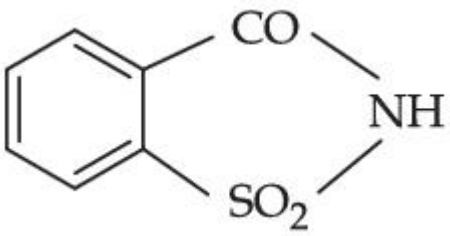
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Q70. MCQ (Single Correct Answer) · **Chemistry**

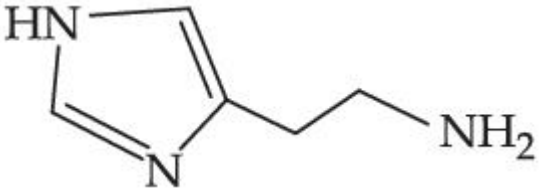
Which one of the following chemicals is responsible for the production of HCl in the stomach leading to irritation and pain?

Open solution

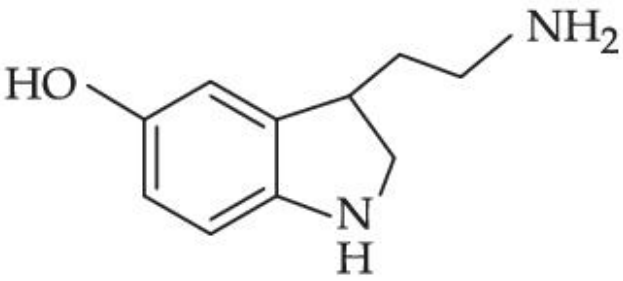
(A)



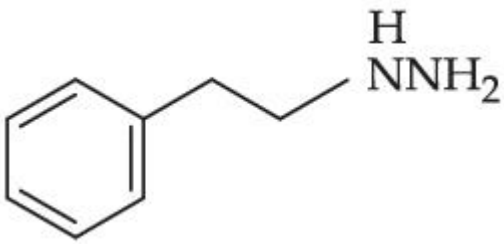
(B)



(C)



(D)



Q71. MCQ (Single Correct Answer) · **Chemistry**

The oxide that gives H_2O_2 most readily on treatment with H_2O is :

(A) PbO_2 (B) Na_2O_2 (C) SnO_2 (D) $BaO_2 . 8H_2O$

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following is used to remove most of plutonium from spent nuclear fuel?

(A) ClF_3 (B) O_2F_2 (C) I_2O_5 (D) BrO_3

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

Lyophilic sols are more stable than lyophobic sols because :

(A) there is a strong electrostatic repulsion between the negatively charged colloidal particles.

(B) the colloidal particles have positive charge.

(C) the colloidal particles have no charge.

(D) the colloidal particles are solvated.

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match List-I with List-II :

	List - I (Name of ore/mineral)		List - II (Chemical formula)
(a)	Calamine	(i)	ZnS
(b)	Malachite	(ii)	$FeCO_3$
(c)	Siderite	(iii)	$ZnCO_3$
(d)	Sphalerite	(iv)	$CuCO_3.Cu(OH)_2$

Choose the most appropriate answer from the options given below :

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(D) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following is formed (mainly) when red phosphorus is heated in a sealed tube at 803 K?

(A) White phosphorus

(B) Yellow phosphorus

(C) β-Black phosphorus

(D) α-Black phosphorus

Open solution

Q76. MCQ (Single Correct Answer) · **Mathematics**

Each of the persons A and B independently tosses three fair coins. The probability that both of them get the same number of heads is :

(A) $\frac{1}{8}$

(B) $\frac{5}{8}$

(C) $\frac{5}{16}$

(D) 1

Open solution

Q77. MCQ (Single Correct Answer) · **Mathematics**

A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point (2, -3) from the line 3x + 4y = 5, is given by :

(A) $10\frac{d^2y}{dx^2} = 11$

(B) $11\frac{d^2x}{dy^2} = 10$

(C) $10\frac{d^2x}{dy^2} = 11$

(D) $11\frac{d^2y}{dx^2} = 10$

Open solution

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Q78. MCQ (Single Correct Answer) · **Mathematics**

The equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to the x-axis is :

(A) $\vec{r}.\left(\hat{j}-3\hat{k}\right)+6=0$ (B) $\vec{r}.\left(\hat{i}+3\hat{k}\right)+6=0$

(C) $\vec{r}.\left(\hat{i}-3\hat{k}\right)+6=0$ (D) $\vec{r}.\left(\hat{j}-3\hat{k}\right)-6=0$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let A(a, 0), B(b, 2b + 1) and C(0, b), $b \neq 0$, $|b| \neq 1$, be points such that the area of triangle ABC is 1 sq. unit, then the sum of all possible values of a is :

(A) $\frac{-2b}{b+1}$ (B) $\frac{2b}{b+1}$ (C) $\frac{2b^2}{b+1}$ (D) $\frac{-2b^2}{b+1}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The Boolean expression $(p \wedge q) \Rightarrow ((r \wedge q) \wedge p)$ is equivalent to :

(A) $(p \wedge q) \Rightarrow (r \wedge q)$ (B) $(q \wedge r) \Rightarrow (p \wedge q)$

(C) $(p \wedge q) \Rightarrow (r \vee q)$ (D) $(p \wedge r) \Rightarrow (p \wedge q)$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Two poles, AB of length a metres and CD of length a + b ($b \neq a$) metres are erected at the same horizontal level with bases at B and D. If BD = x and $\tan\angle ACB = \frac{1}{2}$, then :

(A) $x^2 + 2(a + 2b)x - b(a + b) = 0$ (B) $x^2 + 2(a + 2b)x + a(a + b) = 0$

(C) $x^2 - 2ax + b(a + b) = 0$ (D) $x^2 - 2ax + a(a + b) = 0$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

If $0 < x < 1$ and $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \ldots$, then the value of e^{1+y} at $x = \frac{1}{2}$ is :

(A) $\frac{1}{2}e^2$ (B) 2e (C) $\frac{1}{2}\sqrt{e}$ (D) $2e^2$

Open solution

Q83. Numerical · **Mathematics**

Let S be the sum of all solutions (in radians) of the equation $\sin^4\theta + \cos^4\theta - \sin\theta\cos\theta = 0$ in $[0, 4\pi]$. Then $\frac{8S}{\pi}$ is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane $2x - y + z + 3 = 0$ and let R(3, 5, γ) be a point of this plane. Then the square of the length of the line segment SR is _____.

Open solution

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Q85. Numerical · **Mathematics**

Let A (secθ, 2tanθ) and B (secϕ, 2tanϕ), where $\theta + \phi = \pi/2$, be two points on the hyperbola $2x^2 - y^2 = 2$. If (α, β) is the point of the intersection of the normals to the hyperbola at A and B, then $(2\beta)^2$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

Two circles each of radius 5 units touch each other at the point (1, 2). If the equation of their common tangent is 4x + 3y = 10, and $C_1(\alpha, \beta)$ and $C_2(\gamma, \delta)$, $C_1 \neq C_2$ are their centres, then $|(\alpha + \beta)(\gamma + \delta)|$ is equal to _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

For a transistor α and β are given as $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B}$. Then the correct relation between α and β will be :

(A) $\alpha = \frac{1-\beta}{\beta}$

(B) $\beta = \frac{\alpha}{1-\alpha}$

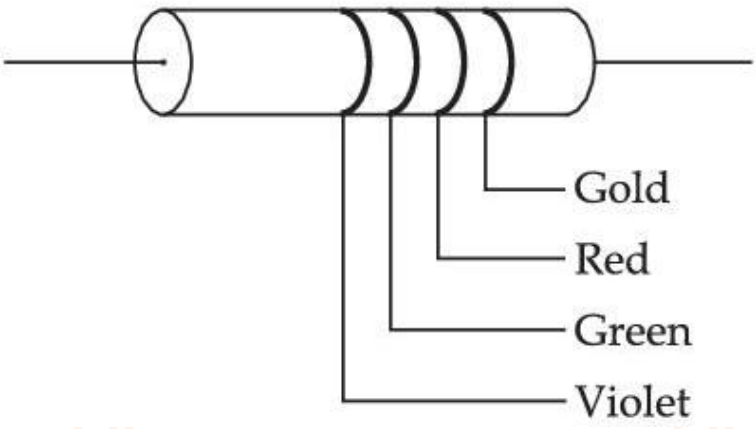
(C) $\alpha\beta = 1$

(D) $\alpha = \frac{\beta}{1-\beta}$

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The Colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is :



(A) $(5700 \pm 285)\ \Omega$

(B) $(7500 \pm 750)\ \Omega$

(C) $(5700 \pm 375)\ \Omega$

(D) $(7500 \pm 375)\ \Omega$

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

An antenna is mounted on a 400 m tall building. What will be the wavelength of signal that can be radiated effectively by the transmission tower upto a range of 44 km?

(A) 37.8 m

(B) 605 m

(C) 75.6 m

(D) 302 m

Open solution

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Q90. Numerical · **Physics**

A heat engine operates between a cold reservoir at temperature T_2 = 400 K and a hot reservoir at temperature T_1. It takes 300 J of heat from the hot reservoir and delivers 240 J of heat to the cold reservoir in a cycle. The minimum temperature of the hot reservoir has to be _____ K.

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Answer Key

Chemistry

1. **B**	2. **C**	3. **A**	4. **D**	5. **A**
6. **A**	7. **B**	8. **D**	9. **D**	10. **B**
11. **D**	12. **D**	13. **16**	14. **2**	15. **84**
16. **19**	17. **271**	18. **760**	19. **50**	20. **6**
21. **3**	22. **964**			

Mathematics

23. **A**	24. **B**	25. **D**	26. **B**	27. **D**
28. **A**	29. **A**	30. **C**	31. **B**	32. **A**
33. **A**	34. **A**	35. **B**	36. **30**	37. **6**
38. **80**	39. **15**	40. **25**	41. **7**	

Physics

42. **B**	43. **D**	44. **D**	45. **C**	46. **C**
47. **A**	48. **C**	49. **D**	50. **D**	51. **B**
52. **B**	53. **A**	54. **A**	55. **A**	56. **C**
57. **D**	58. **B**	59. **1**	60. **15**	61. **20**
62. **40**	63. **400**	64. **1**	65. **9**	66. **2**
67. **34**				

Out of syllabus

68. **C**	69. **C**	70. **B**	71. **B**	72. **B**
73. **D**	74. **A**	75. **D**	76. **C**	77. **D**
78. **A**	79. **D**	80. **A**	81. **C**	82. **A**
83. **56**	84. **72**	85. **Bonus**	86. **40**	87. **B**
88. **D**	89. **B**	90. **500**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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